FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION, NOVEMBER 2024

First Semester

Discipline Specific Core Course-MG1DSCGEO100-THE EARTH: AN INTRODUCTION

OP CODE: 24900120

Answer Scheme

Part A

Attend any 7 of the following in 60 words

Each question carries 2 marks

1. Temperature in the inner core is 5000 - 6000° Celsius (5200°C). The inner core is under intense pressure, which keeps it solid despite high temperatures. Temp. range - 1 mark.
[K] [1]

2. Oceanic crust is denser than continental crust. This is because they are made of denser rocks. Oceanic crust has an average density of 2.9-3.1 g/cm3 (3.0 gm/cm3) while continental crust has an average density of 2.6-2.9 gm/cm3 (2.7gm/cm3). Density range – 1 mark.
[U] [1]

3. A layer of sediment 1 cm or more in thickness. Sedimentary rocks are characterized by bedding. Definition – 1 mark.

[K] [3]

4. Uniformitarianism- the geological processes (weathering, erosion, volcanism, earthquakes, etc.) that occur today, had also occurred in the past. Principle of uniformitarianism propose that 'the present is the key to the past'. Definition- 1 mark.
[U] [2]

5. Strike and dip are measured using a Brunton compass or clinometer. For writing the name of the instruments – 1 mark.
[U] [5]

6.Luster is the light reflected from a mineral surface. Example for a characteristic luster exhibited by a mineral. Definition 1 mark.
[U] [3]

7. Petrology- study of rocks, including their origin, composition, and classification. Branches- Igneous Petrology: The study of rocks formed from solidified magma or lava. Sedimentary Petrology- study of rocks formed from sediment deposition and lithification. Metamorphic Petrology- study of rocks formed from the transformation of preexisting rocks due to heat, pressure, and chemical processes. Definition – ½ mark, Branches - ½ mark.
[U] [1]

8. Pedocal and pedalfer are soil types. Pedocal is formed in arid climate rich in calcium ions, whereas Pedalfer in humid climatic set up characterized by Iron and Aluminum ions. Definition & Description – 1 mark.
[U] [2]

Part B

Write Short Notes on Any 3 of the following in 250 Words

Each question carries 8 marks

9. Describe the internal structure of the Earth-Crust, Mantle and Core. Density, depth and thickness of the layers-6 marks. The internal structure of the Earth is interpreted from seismic discontinuities (2 marks).
[U] [1]

10. Igneous rock – primary rocks formed from magma or lava (1 mark). Origin- extrusive and intrusive igneous rocks with examples (2 mark). Cooling history and grain size or list the major rock forming minerals in an igneous rock- for example -major minerals in granite (1 mark).

Sedimentary rocks- definition (1 mark). Their origin involves the weathering, transportation of the sediments and their deposition in a sedimentary environment, followed by compaction and cementation, characterized by layered nature (bedding) -2 marks. Type of sedimentary rocks (a) clastic rocks and (b) chemically precipitated rocks, including biochemical rocks with examples (1 mark). [U] [3]

11. Folds are warps in rock strata during deformation. Rocks deform when applied stress exceeds their strength. They may deform by ductile flow and bend to form a fold- definition of a fold: 1 mark, origin of a fold: 3 marks. Parts of a fold - limbs, hinge, axial plane, and fold axis- explain each part (4 x 1 = 4 marks) and draw figure- 1 mark.
[U] [5]

12. Fossils- naturally preserved remains of past life. Type of fossils- body fossils, chemical fossils, trace fossils, plant fossils (Definition and description - 4 marks). Fossil animals and plants occur in the geologic record in a definite chronologic order. Fossils provide hints to paleo-environment

conditions. Extinction of an organism reveals the non-adaptable environment conditions existed in the past, for example -extinction of dinosaurs in the Jurassic period (2 marks). The theory of evolution states that life forms have changed/evolved throughout geologic time. The trend of evolution can be studied with the aid of fossils. Interpretations from geologic time scale (2 marks). [U] [4]

13. The processes by which rocks are chemically altered or physically broken into fragments as a result of exposure to atmospheric agents and the pressures and temperatures at or near Earth's surface (Definition -1 Mark). The factors that influence the weathering processes are Rock type, Mineral composition-particle size and shape, Climate, Topography, Time, Biological activity and Role played by geological agents. Explain each terms (7 marks). [U] [2]

Part C

Write Essay on any 1, of the following in 400 Words

Each question carries 12 marks

14. Definition of Fossil -1 mark. Types of Fossils with examples - Body Fossils, Trace Fossils (Ichno fossils), Chemical fossils, Derived /Reworked fossils, Living fossils, Index Fossils, Plant fossils. Explain each - 8 marks. Relevance of index fossils in determining the age of a strata. Fossils provide information about the ages of sedimentary rocks and their depositional environments. Relative age lists the order in which events occurred. For example - Rocks containing dinosaur bones must be younger than those containing trilobite remains. (Description- 3 marks). [U] [4]

15. Earth is a terrestrial planet/ or an introduction (1 mark). Size- 5th largest planet, equatorial radius is 6,378.14 km, polar radius is 6,356.75 km (2 marks). Shape- 1 mark - Oblate spheroid/ Geoid. Volume -1 mark- 260 billion cubic miles. Density- 1 mark - 5.5gm/cm3. Rotation- Definition-1 mark- The Earth's spinning motion on its axis, which takes 24 hours. Revolution- Definition- 1 mark- The Earth's motion around the Sun in an elliptical path, which takes 365 days and 6 hours.

Earth's rotation and revolution significantly influence its climate. The rotation of the Earth on its axis causes the cycle of day and night, with the side facing the sun experiencing day (and warmth) and the side away from the sun experiencing night (and cooler temperatures). The revolution of the Earth around the sun, combined with the tilt of the Earth's axis, causes the changing seasons. When the Earth's axis tilts towards the sun, it is summer for that hemisphere because the sun's rays hit the Earth at a more direct angle, causing warmer temperatures. When the axis tilts away from the sun, it is winter for that hemisphere because the sun's rays hit the Earth at a less direct angle, causing cooler temperatures (4 mark). [U] [1]